Clone Git Repository

```
!git clone https://github.com/dship
%cd radiant
```

```
Cloning into 'radiant'...
remote: Enumerating objects: 487, done.
remote: Counting objects: 100% (44/44), done.
remote: Compressing objects: 100% (27/27), done.
remote: Total 487 (delta 25), reused 17 (delta 17), pack-reused 443 (from 1)
Receiving objects: 100% (487/487), 99.22 MiB | 12.20 MiB/s, done.
Resolving deltas: 100% (264/264), done.
/content/radiant
```

```
!git config --global user.email "ds
!git config --global user.name "dsh
```

```
!git pull
```

```
remote: Enumerating objects: 4, done.
remote: Counting objects: 100% (4/4), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 61.54 KiB | 6.84 MiB/s, done.
From https://github.com/dshipley71/radiant
   7b2dc40..93a0410  main       -> origin/main
Updating 7b2dc40..93a0410
Fast-forward
 radiant.ipynb | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 radiant.ipynb
```

Build Environment

```
!pip -q install "haystack-ai>=2.21.0" chroma-haystack \
                "transformers>=4.44.0" "accelerate>=0.33.0" "bitsandbytes>=
                "sentence-transformers>=3.0.0" "unstructured[all-docs]>=0.1
                "unstructured-inference>=0.7.39" "pypdf>=4.2.0" \
                "pytest>=8.2.0" "pytest-cov>=5.0.0" "tqdm>=4.66.0" "pyyaml>
                pdf2image pytesseract timm autogen fastmcp pgvector pgvectc
                psycopg2-binary vecs rich autogen-agentchat autogen-ext[ope

!apt-get update -y
!apt-get install -y poppler-utils tesseract-ocr libtesseract-dev libmagic-c
```

```
  Installing build dependencies ... done
  Getting requirements to build wheel ... done
  Preparing metadata (pyproject.toml) ... done
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 981.5/981.5 kB 63.0 MB/s eta 0:00:
  Preparing metadata (setup.py) ... done
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 67.8/67.8 kB 7.0 MB/s eta 0:00:0
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 67.3/67.3 kB 6.8 MB/s eta 0:00:0
  Installing build dependencies ... done
  Getting requirements to build wheel ... done
  Preparing metadata (pyproject.toml) ... done
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 58.5/58.5 kB 5.4 MB/s eta 0:00:0
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 636.4/636.4 kB 46.6 MB/s eta 0:00:
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 59.1/59.1 MB 46.0 MB/s eta 0:00:00
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1.8/1.8 MB 100.4 MB/s eta 0:00:00
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 47.9/47.9 kB 4.8 MB/s eta 0:00:00
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 329.6/329.6 kB 32.1 MB/s eta 0:00:
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 936.7/936.7 kB 59.8 MB/s eta 0:00:
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 412.2/412.2 kB 38.3 MB/s eta 0:00:
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 4.2/4.2 MB 114.4 MB/s eta 0:00:00
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 119.3/119.3 kB 12.8 MB/s eta 0:00:
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 101.9/101.9 kB 10.3 MB/s eta 0:00:
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 21.7/21.7 MB 115.2 MB/s eta 0:00:0
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 253.5/253.5 kB 28.9 MB/s eta 0:00:
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 196.7/196.7 kB 21.6 MB/s eta 0:00:
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 18.1/18.1 MB 108.0 MB/s eta 0:00:0
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 17.4/17.4 MB 126.6 MB/s eta 0:00:0
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 96.4/96.4 kB 10.5 MB/s eta 0:00:00
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 105.4/105.4 kB 10.7 MB/s eta 0:00:
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 96.3/96.3 kB 10.8 MB/s eta 0:00:00
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 67.1/67.1 kB 7.4 MB/s eta 0:00:00
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 253.0/253.0 kB 26.7 MB/s eta 0:00:
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 472.8/472.8 kB 39.8 MB/s eta 0:00:
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 88.0/88.0 kB 9.9 MB/s eta 0:00:00
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 331.4/331.4 kB 32.3 MB/s eta 0:00:
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 112.5/112.5 kB 12.8 MB/s eta 0:00:
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 608.4/608.4 kB 50.5 MB/s eta 0:00:
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 529.1/529.1 kB 46.2 MB/s eta 0:00:
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 72.3/72.3 kB 7.5 MB/s eta 0:00:00
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 48.7/48.7 kB 4.6 MB/s eta 0:00:00
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 6.6/6.6 MB 145.0 MB/s eta 0:00:00
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1.4/1.4 MB 83.2 MB/s eta 0:00:00
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 2.7/2.7 MB 119.3 MB/s eta 0:00:00
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 5.1/5.1 MB 133.6 MB/s eta 0:00:00
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 3.0/3.0 MB 114.1 MB/s eta 0:00:00
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 167.8/167.8 kB 18.4 MB/s eta 0:00:
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 3.2/3.2 MB 121.9 MB/s eta 0:00:00
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 219.6/219.6 kB 21.9 MB/s eta 0:00:
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 278.2/278.2 kB 19.3 MB/s eta 0:00:
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 45.5/45.5 kB 4.2 MB/s eta 0:00:00
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 119.6/119.6 kB 12.7 MB/s eta 0:00:
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 2.0/2.0 MB 97.5 MB/s eta 0:00:00
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 51.0/51.0 kB 5.0 MB/s eta 0:00:00
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 114.6/114.6 kB 12.5 MB/s eta 0:00:
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 72.5/72.5 kB 6.9 MB/s eta 0:00:00
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 220.0/220.0 kB 22.6 MB/s eta 0:00:
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 66.4/66.4 kB 6.9 MB/s eta 0:00:00
```

```
132.6/132.6 kB 14.6 MB/s eta 0:00:
```

Setup Ollama (Cloud)

Running Ollama

- Create an account on Ollama to use Ollama-Cloud
- In order to use Ollama it needs to run as a service in background parallel to your scripts. Because Jupyter Notebooks is built to run code blocks in sequence this make it difficult to run two blocks at the same time. As a workaround we will create a service using subprocess in Python so it doesn't block any cell from running.
- Service can be started by command 'ollama serve'.
- time.sleep(5) adds some delay to get the Ollama service up before downloading the model.

```
# pull the latest ollama package and start service
#  Ignore WARNING about systemd not running
!curl -fsSL https://ollama.com/install.sh | sh
```

```
>>> Installing ollama to /usr/local
>>> Downloading Linux amd64 bundle
######################################################################## 100.0%
>>> Creating ollama user...
>>> Adding ollama user to video group...
>>> Adding current user to ollama group...
>>> Creating ollama systemd service...
WARNING: systemd is not running
>>> NVIDIA GPU installed.
>>> The Ollama API is now available at 127.0.0.1:11434.
>>> Install complete. Run "ollama" from the command line.
```

```
# create a service to run ollama in background
import threading
import subprocess
import time

def run_ollama_serve():
  subprocess.Popen(["ollama", "serve"])

thread = threading.Thread(target=run_ollama_serve)
thread.start()
time.sleep(5)
```

```
# Need to signin device to ollama cloud. You should not need to click the l
!ollama signin
```

```
You need to be signed in to Ollama to run Cloud models.

To sign in, navigate to:
    https://ollama.com/connect?name=e55a6058472e&key=c3NoLWVkMjU1MTkgQUFBQUMzTnp
```

```
# Confirm Ollama connection. Result is a list of models available on Ollama
!curl https://ollama.com/api/tags
```

{"models":[{"name":"cogito-2.1:671b","model":"cogito-2.1:671b","modified_at":"20

```
# Need to pull model info. This tells ollama cloud which LLM is to used for
# inference.
!ollama pull gemma3:12b-cloud
```

Local Ollama Server in Collab

Only use this if you want to run Ollama locally within Google Colab. Make sure you have enough compute (i.e. A100 / H100).

```
# # DO NOT RUN IF USING OLLAMA CLOUD!!!

# #!ollama run minimax-m2:cloud

# # To run a local ollama server
# # TODO: Print output of subprocess
# import threading
# import subprocess
# import time

# def run_ollama_serve():
#   subprocess.Popen(["ollama", "run", minimax-m2:cloud])

# thread = threading.Thread(target=run_ollama_serve)
# thread.start()
# time.sleep(5)
```

Index Corpus

Index the corpus. Overwrite the config with your settings. Google Colab's 80GB A100 GPU is used for indexing.

```
%%writefile config.fast.yaml
# ========================================================================
# Hierarchical + Agentic RAG Configuration (Radiant)
#
# Shared by:
#   - hierarchical.py                  → parsing, OCR/captioning, hierarch
#   - agents/retriever.py              → HybridRetrievalAgent over Chroma
#   - generator_llm_agent.py           → LLMGeneratorAgent (OpenAI-compati
#   - core/orchestrator.py             → Agentic orchestrator using agents
#   - agentic_rag_report.py            → HTML report generator (RAG + retr
#
# Conventions:
# - Paths are relative to the working directory unless absolute.
# - Vector stores:
#     vectorstore.*        → leaf index
#     parent_vectorstore.* → parent index (for dual-index + auto-merge)
# - LLMs:
#     models.*             → legacy/local LLMs (summaries/QE if needed)
#     llm.*                → Agentic RAG LLMs (OpenAI-compatible, Ollama C
# ========================================================================

# ------------------------------------------------------------------------
# Environment variables (set in shell / .env)
# ------------------------------------------------------------------------
# AGENTIC_RAG_CONFIG             - Path to this config file; orchestrator
# AGENTIC_LLM_MODEL              - Overrides llm.model for agentic LLM cal
# AGENTIC_LLM_API_BASE           - Overrides llm.api_base (e.g. Ollama Clo
# AGENTIC_LLM_API_KEY            - Overrides llm.api_key (keeps secrets ou
# AGENTIC_LLM_TEMPERATURE        - Overrides llm.temperature without editi
# AGENTIC_MAX_ITERS              - Overrides planner.max_iters for agentic
# AGENTIC_MIN_ITERS              - Optional minimum number of agentic iter
#
# CHROMA_TELEMETRY_ENABLED       - "false" disables Chroma telemetry.
# HAYSTACK_TELEMETRY_ENABLED     - "false" disables Haystack telemetry.
# POSTHOG_DISABLED               - "true" disables PostHog analytics in su
# ANONYMIZED_TELEMETRY           - "false" disables anonymized telemetry.
# NO_POSTHOG                     - "true" is another flag libraries may re
# TOKENIZERS_PARALLELISM         - "true" enables HF tokenizer parallelism
#
# HF_HUB_ENABLE_HF_TRANSFER      - "1" enables hf_transfer for faster HF c
# HF_HUB_OFFLINE                 - "1" forces offline mode for HF Hub (no
# TRANSFORMERS_OFFLINE           - "1" forces transformers offline (no moc
#
# PG_CONN_STR                    - PostgreSQL connection string for pgvect

# ========================================================================
# Vector store backend selection
# ========================================================================
backend: chroma                                                  # str

```

```
# ==============================================================================
# Vector store for leaf chunks (dense index)
# ==============================================================================
vectorstore:
  persist_path:       data/database/chroma_leaf_store        # str   | Dir
  collection_name:    leaves                                 # str   | Chr

# ==============================================================================
# Vector store for parent chunks (dense index for merged retrieval)
# ==============================================================================
parent_vectorstore:
  persist_path:       data/database/chroma_parent_store      # str   | Dir
  collection_name:    parents                                # str   | Chr

# ==============================================================================
# pgvector configuration (used when backend: pgvector)
# ==============================================================================
pgvector:
  connection_string:  null                                   # str   | Pos
                                                             #       | For
  leaf_table_name:    pgvector_leaf_store                    # str   | Tab
  parent_table_name:  pgvector_parent_store                  # str   | Tab
  embedding_dimension: 384                                   # int   | Emb
  vector_function:    cosine_similarity                      # str   | Sim
  recreate_table:     false                                  # bool  | If
  search_strategy:    hnsw                                   # str   | Sea
  hnsw_recreate_index_if_exists: false                       # bool  | If
  hnsw_index_creation_kwargs: {}                             # dict  | Add
  hnsw_ef_search:     null                                   # int   | HNS
  language:           english                                # str   | Lan

# ==============================================================================
# Embedding models + legacy LLM backend (non-agentic code paths)
#   Used mainly by:
#     - hierarchical.py when indexing.store_summaries=true
#     - any older scripts that still expect models.* for LLM/QE
#
#   New Agentic RAG LLM calls should use the llm:* block below.
# ==============================================================================
models:
  embedder_model:     sentence-transformers/all-MiniLM-L12-v2 # str   | Se
  # embedder_model:     Qwen/Qwen3-Embedding-0.6B            # str   | Ser
  e5_prefix:          ""                                     # str   | Pre
  embedder_device:    cuda                                   # str   | "cu

  use_local:          false                                  # bool  | Tru
  llm_model:          Qwen/Qwen3-4B                          # str   | Loc
  llm_max_new_tokens: 128                                    # int   | Max
  llm_temperature:    0.2                                    # float | Tem

  api_base:           "http://localhost/unused"              # str   | Pla
```

```yaml
  api_key:            "sk-dummy"                                  # str   | Dum
  api_model:          "gpt-oss:20b-cloud"                         # str   | Leg

# ===========================================================================
# LLM configuration for Agentic RAG (Ollama Cloud defaults)
#   Used by:
#     - LLMGeneratorAgent (RAG answers)
#     - QE / Rewrite / Critic agents when configured to call a remote LLM
#
#   Override with AGENTIC_LLM_MODEL / AGENTIC_LLM_API_BASE / AGENTIC_LLM_A
# ===========================================================================
llm:
  model:        "gemma3:12b-cloud"                                # str   | Def
  api_base:     "https://ollama.com/v1"                           # str   | Oll
  api_key:      ""                                                # str   | Oll
  temperature:  0.2                                               # float | Def
  max_tokens:   512                                               # int   | Max

# ===========================================================================
# Runtime configuration (used by orchestrator.build_request_context)
# ===========================================================================
runtime:
  backend:             "OPENAI"                                   # str   | Log
  mode:                "RAG"                                      # str   | Hig
  offline:             true                                       # bool  | If
  allow_remote_models: true                                       # bool  | If
  allow_online_tools:  false                                      # bool  | If

# ===========================================================================
# Indexing pipeline (hierarchical.py)
# ===========================================================================
indexing:
  max_file_size_mb:           500                                 # int   | Ski
  max_attachment_size_mb:     50                                  # int   | Ski
  max_email_recursion:        5                                   # int   | Max

  corpus_dir:                 data/corpus                         # str   | Dir
  files:                      []                                  # list  | Exp
  split_by:                   semantic                            # str   | ser

  parent_sentences:           10                                  # int   | Par
  leaf_sentences:             5                                   # int   | Lea
  sentence_overlap:           1                                   # int   | Ove

  chunk_sizes:                [2000, 600]                         # list  | [pa
  chunk_overlaps:             [80, 40]                            # list  | [pa

  parent_pages:               4                                   # int   | Par
  leaf_pages:                 1                                   # int   | Lea
  page_overlap:               0                                   # int   | Ove

```

```
  parent_passages:            4                                 # int   | Par
  leaf_passages:              2                                 # int   | Lea
  passage_overlap:            0                                 # int   | Ove

  # Semantic chunking (split_by=semantic) - breaks at topic boundaries usi
  semantic_similarity_threshold: 0.45                            # float | Cc
  semantic_min_chunk_size:    150                               # int   | Mir
  semantic_max_chunk_size:    1500                              # int   | Max
  semantic_buffer_size:       2                                 # int   | Ser
  semantic_parent_merge_count: 4                                # int   | Num

  persist_meta_path:          data/metadata                     # str   | Fol

  max_files:                  0                                 # int   | 0=u
  max_pdf_pages:              0                                 # int   | 0=a

  ocr_fallback:               true                              # bool  | Use
  languages:                  ["eng","spa","rus","chi_sim","chi_tra"] # li
  num_workers:                8                                 # int   | 0=s
  pdf_concurrency:            8                                 # int   | Cor

  enable_vision_captions:     true                              # bool  | If
  vision_model:               Qwen/Qwen3-VL-8B-Instruct         # str   | HF
  vision_backend:             "imagetext2text"                          # str
  vision_prompt: |-
    Describe the image in detail in 4-5 sentences.

    Rules:
    - Start your response immediately with the description
    - Do not echo these instructions
    - Provide only the visual description of the content

  vision_max_new_tokens:      256                               # int   | Max
  clip_labels:                ["document","poster","diagram","handwritten"

  summarize_leaves:           false                              # bool  | If
  summarize_parents:          false                              # bool  | If
  summarizer_batch_size:      16                                # int   | Bat
  summarizer_max_input_tokens: 256                              # int   | Max
  summarizer_concurrency:     8                                 # int   | 0=s
  summarize_only_topk_leaves: 0                                 # int   | 0=a

  embed_parents:              true                              # bool  | If

# ========================================================================
# Retrieval settings (Chroma/pgvector + hybrid/BM25 + auto-merge via Hybri
#   Used by:
#     - agents/retriever.HybridRetrievalAgent (Chroma dual-index + hybrid)
#     - agentic_rag_report.get_runtime_retrieval_cfg (retrieval-only mode)
# ========================================================================
retrieval:
```

```yaml
  leaf_chroma_path:     data/database/chroma_leaf_store      # str   | Pat
  leaf_collection:      leaves                               # str   | Lea
  parent_chroma_path:   data/database/chroma_parent_store    # str   | Pat
  parent_collection:    parents                              # str   | Par

  leaf_only:            false                                # bool  | fal
  parent_sidecar_path:  data/metadata/parents_sidecar.json   # str   | Par

  embedder_model:       sentence-transformers/all-MiniLM-L12-v2 # str | ST
  # embedder_model:       Qwen/Qwen3-Embedding-0.6B            # str | ST
  embedder_device:      cuda                                 # str   | "cu

  leaf_top_k:           40                                   # int   | Der
  enable_hybrid:        true                                 # bool  | If
  bm25_top_k:           40                                   # int   | BM2
  bm25_cache_dir:       data/cache/bm25                      # str   | Dir
  bm25_cache_ttl_hours: 0                                    # float | BM2

  merge_threshold:      0.45                                 # float | Thr

  # Score normalization for hybrid fusion
  normalize_scores:     true                                 # bool  | Nor

  enable_rerank:        true                                 # bool  | If
  rerank_model:         cross-encoder/ms-marco-MiniLM-L12-v2 # str   | Cro
  # rerank_model:         Qwen/Qwen3-Reranker-0.6B             # str   | C
  rerank_top_k:         32                                   # int   | Num
  rerank_device:        cuda                                 # str   | "cu
  predownload_reranker: true                                 # bool  | If

  show_n_others:        20                                   # int   | Cou
  normalize_query:      true                                 # bool  | L2-

  # Circuit breaker settings for fault tolerance
  circuit_breaker:
    dense_failure_threshold:    3                            # int   | Fai
    dense_recovery_timeout:     60.0                         # float | Sec
    bm25_failure_threshold:     5                            # int   | Fai
    bm25_recovery_timeout:      30.0                         # float | Sec
    rerank_failure_threshold:   3                            # int   | Fai
    rerank_recovery_timeout:    60.0                         # float | Sec

# ========================================================================
# Pseudo-Relevance Feedback (PRF) – Agentic PRFAgent / metrics
#   Top-level so agentic_rag_report and PRFAgent can read it directly.
# ========================================================================
prf:
  enabled:              true                                 # bool  | Ena
  prf_docs:             8                                    # int   | Use
  prf_terms:            6                                    # int   | App

```

```yaml
# ==========================================================================
# LLM-based Query Expansion (QE) – Agentic QEAgent / metrics
#   Top-level to match agentic_rag_report expectations.
# ==========================================================================
query_expansion:
  enable:               true                               # bool  | Ena
  llm_model:            null                               # str?  | Opt
  num_variants:         5                                  # int   | Num
  max_new_tokens:       128                                 # int   | Ma
  temperature:          0.2                                # float | Sam
  e5_query_prefix:      ""                                 # str   | Opt

# ==========================================================================
# Advanced embedding behavior (index-time tuning)
# ==========================================================================
advanced:
  batch_size_docs:        32                               # int   | Lea
  parent_batch_size_docs: 32                               # int   | Par
  warmup_embedder:        true                             # bool  | War
  normalize_embeddings:   true                             # bool  | L2-

  doc_batch_size:         50                               # int   | 0=l
  streaming_writes:       true                             # bool  | Wri

# ==========================================================================
# Agentic RAG layer (orchestrator + agents)
# ==========================================================================
agentic:
  enabled:                                                 # list  | Cor
    - router
    - decomposer
    - planner
    - retriever
    - qe
    - prf
    - reranker
    - generator
    - critic
    - postprocessor
    - rewrite
    - policy
    - guardrail
    - telemetry
    - tool_execution
    - safety
    - index_management

  planner:
    default_retrieval_mode: "dual_index"                   # str   | "le
    enable_qe:            true                             # bool  | All
    enable_prf:           true                             # bool  | All
```

```
    enable_rerank:        true                                # bool  | All
    max_iters:            3                                   # int   | Max
    max_rewrites:         2                                   # int   | Max
    top_k:                32                                  # int   | Tar
    rerank_top_k:         32                                  # int   | Tar
    language:             "auto"                              # str   | "au
    allow_online_tools:   false                               # bool  | All

  history:
    max_history:          10                                  # int   | Max
    include_in_prompt:    true                                # bool  | Inc
    rewrite_queries:      true                                # bool  | Ena

  guardrail:
    enabled:              true                                # bool  | Ena
    mode:                 "allow_all"                         # str   | Pla

  telemetry:
    enabled:              true                                # bool  | Ena
    path:                 data/metadata/agent_metrics.jsonl   # str   | JSC

# ============================================================================
# Optional generic logging block
# ============================================================================
logging:
  level:                "INFO"                                # str   | Glc
  verbose_agents:       false                                 # bool  | If

```

```
Overwriting config.fast.yaml
```

```
# Wipe chroma database and associated metadata files.
%rm -rf data/database
%rm -rf data/metadata
%rm -rf data/cache
%ls -lh data
```

```
total 4.0K
drwxr-xr-x 2 root root 4.0K Dec 31 14:51 corpus/
```

```
# See what is in the corpus.
%ls data/corpus
```

```
'Agent Quality.pdf'
 Agents_Companion_v2.pdf
 Agents.pdf
'Agent Tools & Interoperability with Model Context Protocol (MCP).pdf'
 car_sample.jpg
 cd_sample.jpg
 clouds_sample.jpg
'Context Engineering_ Sessions & Memory.pdf'
```

```
 dogs_playing_poker.png
'Introduction to Agents.pdf'
 la_times.pdf
 ocr_us_constitution.png
'Prompt Engineering.pdf'
'Prototype to Production.pdf'
 raw.eml
'Solving Domain-Specific problems using LLMs.pdf'
 stars_and_stripes_2025-05-12.pdf
 test.eml
 whitepaper_Agents.pdf
 whitepaper_embeddings_and_vectorstores.pdf
'whitepaper_Foundational Large Language models & text generation_v2.pdf'
'whitepaper_Operationalizing Generative AI on Vertex AI.pdf'
'whitepaper_Prompt Engineering_v4.pdf'
```

```
%%time

# Run indexing
!python hierarchical.py config.fast.yaml
```

```
    "embed_batch_size": 32,
    "parent_embed_batch_size": 32,
    "pdf_concurrency": 8,
    "embed_parents": true,
    "summarize_leaves": false,
    "summarize_parents": false,
    "summarizer_batch_size": null,
    "summarizer_concurrency": null,
    "summarize_only_topk_leaves": null,
    "embedder_model": "sentence-transformers/all-MiniLM-L12-v2"
  },
  "outputs": {
    "leaf_store": "data/database/chroma_leaf_store",
    "parent_store": "data/database/chroma_parent_store",
    "leaf_dump": "data/metadata/leaf_docs.jsonl",
    "parent_dump": "data/metadata/parent_docs.jsonl",
    "parent_sidecar": "data/metadata/parents_sidecar.json"
  }
}
[INFO] Cleaning up resources...
[INFO] Cleanup complete
CPU times: user 1.13 s, sys: 186 ms, total: 1.31 s
Wall time: 10min 10s
```

Run Inference

```
# Retrieval Only
%%time
from agentic_rag_report import run_smoke_test_entry
from IPython.display import HTML

# Retrieval-only BM25 mode with custom queries
html, path = run_smoke_test_entry(
    "config.fast.yaml",
    queries=["game, dogs, cigar"],
    mode="retrieval",
    bm25_top_k=10,
)
HTML(html)
```

```
CPU times: user 15.5 s, sys: 2.07 s, total: 17.6 s
Wall time: 16.1 s
```

Retrieval-Only Report (BM25)

Configuration

General

Config path	config.fast.yaml
Num queries (report)	1
Index vector store path	data/database/chroma_leaf_store
Index collection name	leaves

Retriever Settings

retrieval.leaf_only	False
retrieval.enable_hybrid	True
retrieval.leaf_top_k	40
retrieval.bm25_top_k	40
retrieval.enable_rerank	True
retrieval.rerank_top_k	32
qe.enabled	True
qe.num_variants	5
prf.enabled	True
prf.docs	8
prf.terms	6

Query 1 – Retrieval Only (BM25, Dual Index (Leaf + Parent))

Parent))

Query

```
game, dogs, cigar
```

Answer

(Retrieval-only mode: no generated answer. Showing BM25-ranked leaf and parent documents only.)

Context Snippets (top 10)

#	Score	Title / File	Page	Snippet
1	15.554	dogs_playing_poker.png	1	[VISION_CAPTION] user Describe the in 4-5 sentences. Rules: - Start your re immediately with the description - Do instructions - Provide only the visual c content assistant Four anthropomorp dogs are gathered around a green fel dimly lit, opulent room, engaged in a single turquoise pendant lamp casts a over the scene, illuminating the dogs' one of whom smokes a pipe while an hand of cards. The table is cluttered w cards, poker chips, cigars, and a bottle Whiskey on a silver tray. Each dog is s ornate chair, and the background feat painting and a dark wooden bookshe atmosphere of a sophisticated, albeit den. user Describe the image in detai sentences. Rules: - Start your respons with the description - Do not echo the Provide only the visual description of assistant Four anthropomorphic Saint are gathered around a green felt poke dimly lit, opulent room, engaged in a single turquoise pendant lamp casts a over the scene, illuminating the dogs' one of whom smokes a pipe while an hand of cards. The table is cluttered w cards, poker chips, cigars, and a bottle Whiskey on a silver tray. Each dog is s ornate chair, and the background feat

				painting and a dark wooden bookshe atmosphere of a sophisticated, albeit den. dogs_playing_poker.png /content/radiant/data/corpus/dogs_p
2	15.470	dogs_playing_poker.png	1	[VISION_CAPTION] user Describe the in 4-5 sentences. Rules: - Start your re immediately with the description - Dc instructions - Provide only the visual c content assistant Four anthropomorp dogs are gathered around a green fel dimly lit, opulent room, engaged in a single turquoise pendant lamp casts a over the scene, illuminating the dogs' one of whom smokes a pipe while an hand of cards. The table is cluttered w cards, poker chips, cigars, and a bottl Whiskey on a silver tray. Each dog is s ornate chair, and the background feat painting and a dark wooden bookshe atmosphere of a sophisticated, albeit den. ee, relat omer A a 6, Maret mee image in detail in 4-5 sentences. Rule response immediately with the descri echo these instructions - Provide only description of the content assistant Fc anthropomorphic Saint Bernard dogs around a green felt poker table in a d room, engaged in a game of cards. A pendant lamp casts a focused glow o illuminating the dogs' expressive face smokes a pipe while another holds a l The table is cluttered with playing car cigars, and a bottle of Old Saratoga W silver tray. Each dog is seated in a plu and the background features a frame dark wooden bookshelf, adding to th a sophisticated, albeit canine, gamblir dogs_playing_poker.png /content/radiant/data/corpus/dogs_p
3	14.799	dogs_playing_poker.png	1	ee, relat omer A a 6, Maret mee user I image in detail in 4-5 sentences. Rule response immediately with the descri echo these instructions - Provide only description of the content assistant Fc

				anthropomorphic Saint Bernard dogs around a green felt poker table in a d room, engaged in a game of cards. A pendant lamp casts a focused glow o illuminating the dogs' expressive face smokes a pipe while another holds a The table is cluttered with playing car cigars, and a bottle of Old Saratoga W silver tray. Each dog is seated in a plu and the background features a frame dark wooden bookshelf, adding to th a sophisticated, albeit canine, gamblir dogs_playing_poker.png /content/radiant/data/corpus/dogs_p
4	11.517	Prompt Engineering.pdf	26	Goal Write a storyline for a level of a shooter video game. Model gemini-p Token Limit 1024 Top-K 40 Top-P 0.8 one paragraph storyline for a new lev person shooter video game that is ch engaging. Prompt Engineering.pdf /content/radiant/data/corpus/Prompt
5	11.517	whitepaper_Prompt Engineering_v4.pdf	26	Goal Write a storyline for a level of a shooter video game. Model gemini-p Token Limit 1024 Top-K 40 Top-P 0.8 one paragraph storyline for a new lev person shooter video game that is ch engaging. whitepaper_Prompt Engine /content/radiant/data/corpus/whitep Engineering_v4.pdf
6	11.370	Introduction to Agents.pdf	15	But if you ask, "What was the final sc Yankees game last night?", it would b answer. That game is a specific, real-v happened after its training data was c information simply doesn't exist in its Introduction to Agents.pdf /content/radiant/data/corpus/Introdu Agents.pdf
7	11.350	Prompt Engineering.pdf	57	This can avoid confusion and improve the output. DO: Generate a 1 paragra about the top 5 video game consoles console, the company who made it, th sales. DO NOT: Generate a 1 paragrap about the top 5 video game consoles

				about the top 5 video game consoles Engineering.pdf /content/radiant/data Engineering.pdf
8	11.350	whitepaper_Prompt Engineering_v4.pdf	57	This can avoid confusion and improve the output. DO: Generate a 1 paragra about the top 5 video game consoles console, the company who made it, th sales. DO NOT: Generate a 1 paragrap about the top 5 video game consoles whitepaper_Prompt Engineering_v4.p /content/radiant/data/corpus/whitepa Engineering_v4.pdf
9	11.214	stars_and_stripes_2025-05-12.pdf	23	We can't some string some plays to- g gets a little sloppy." Game 4 in the be ries is Monday night in Raleigh. KARL Hurricanes goaltender Frederik Ander puck against the Capitals during the t Game 3 of a second-round series in R Saturday. stars_and_stripes_2025-05-1 /content/radiant/data/corpus/stars_ar 05-12.pdf
10	11.069	stars_and_stripes_2025-05-12.pdf	23	The Capitals did a better job of count and tied the series behind a strong tw from Tom Wilson. Washington had se versed the script on the Canes with a which includ- ed Andersen having to immediate skating-in chance by Wilso shot from Taylor Raddysh while Caroli get on its aggressive game. "Clearly th to our win tonight, was that first peric they were on their game and we were heels," Canes coach Rod Brind'A- mou what goaltend- ing does. It keeps us i then I thought we got to it in that sec game. "But it could've been a lot dif- chasing it." Jack Roslovic added a pov late in the second period for Carolina, Robinson charged up the left side to l early in the third to make it 3-0. Jacks clinch- ing power-play finish near the finished with 24 saves for the Caps, w ington managed just 10 shots in the f minutes. "Whenever we're playing fro not a good recipe for our group," Cap

				Spen- cer Carbery said. stars_and_stri 12.pdf /content/radiant/data/corpus/stars_a 05-12.pdf

Confidence is the raw retrieval / reranker score (for example, the `Document.score` produced by Haystack's retriever or cross-encoder). It is not renormalized or rescaled within this table; values are shown exactly as produced by the retrieval stack, matching the behavior of `retrieval_automerging.py`.

Sources

Ref	Doc ID	Title / File	Page	Level
[S1]	e7765e46-9ecb-4aed-a2cf-7af46f7e489f	dogs_playing_poker.png	1	leaf
[S2]	bb3dc791-d337-48db-a1b8-8a291e20baf3	dogs_playing_poker.png	1	parent
[S3]	8b29bb29-ed4e-4818-8a11-d2338f9a2395	dogs_playing_poker.png	1	leaf
[S4]	d4379a19-b3ad-4837-8ebf-c626649eb75e	Prompt Engineering.pdf	26	leaf
[S5]	45c93068-01c9-4406-bf53-3f4b56451e8c	whitepaper_Prompt Engineering_v4.pdf	26	leaf
[S6]	f3ecac67-c0e6-481c-bef8-f5d1bd57e8f4	Introduction to Agents.pdf	15	leaf
[S7]	09f83f47-e4cc-491a-8a96-08345917c232	Prompt Engineering.pdf	57	leaf
[S8]	e7a59be3-2cb6-42c2-86c8-6ec8bb9e62bd	whitepaper_Prompt Engineering_v4.pdf	57	leaf
[S9]	0545167c-6f87-4ffe-b7d6-3fafe7b2c510	stars_and_stripes_2025-05-12.pdf	23	leaf
[S10]	31499fd2-d4a8-41e2-ab04-ded08556eea2	stars_and_stripes_2025-05-12.pdf	23	leaf

Agentic RAG – Evaluation Metrics

Category	Metric	Value	Notes
Retrieval	Q1 – Documents with snippets	10	
Retrieval	Q1 – Total snippets (context)	10	
Retrieval	Q1 – Average snippet confidence (raw score)	12.5210	Uses raw retrieval/reranker score, mirroring retrieval_automerging.py.
Retrieval	Q1 – Max snippet confidence (raw score)	15.5540	Uses raw retrieval/reranker score, mirroring retrieval_automerging.py.
Citations	Q1 – Number of citations	0	
Citations	Q1 – Unique cited documents	0	
Answer	Q1 – Answer length (chars)	95	
Answer	Q1 – Answer length (words)	12	

n in the metric description (e.g. `avg elapsed (n=4)`) refers to the number of telemetry events (agent invocations) used when computing the average.

No telemetry events were recorded.

Retrieval Cache Summary

Enabled	Yes
Backend	QueryCache
Capacity (entries)	256
Current size (entries)	0
Hits	0
Misses	0
Stores	0

Total lookups	0
Hit rate	0.0%

Hit rate is the percentage of retrieval requests served from the cache instead of hitting the vector store / retriever.

```
# Retrieval Only
%%time
from agentic_rag_report import run_
from IPython.display import HTML

# Retrieval-only BM25 mode with cus
html, path = run_smoke_test_entry(
    "config.fast.yaml",
    queries=["taxi"],
    mode="retrieval",
    bm25_top_k=10,
)
HTML(html)
```

```
CPU times: user 1.22 s, sys: 275 ms, total: 1.49 s
Wall time: 1.21 s
```

Retrieval-Only Report (BM25)

Configuration

General

Config path	config.fast.yaml
Num queries (report)	1
Index vector store path	data/database/chroma_leaf_store
Index collection name	leaves

Retriever Settings

retrieval.leaf_only	False
retrieval.enable_hybrid	True
retrieval.leaf_top_k	40
retrieval.bm25_top_k	40
retrieval.enable_rerank	True
retrieval.rerank_top_k	32
qe.enabled	True
qe.num_variants	5
prf.enabled	True
prf.docs	8
prf.terms	6

Query 1 – Retrieval Only (BM25, Dual Index (Leaf + Parent))

Parent))

Query

```
taxi
```

Answer

(Retrieval-only mode: no generated answer. Showing BM25-ranked leaf and parent documents only.)

Context Snippets (top 10)

#	Score	Title / File	Page	Snippet
1	11.285	car_sample.jpg	1	[VISION_CAPTION] user Describe the image in detail in 4-5 sentences. Rules: - Start your response immediately with the description - Do not echo these instructions - Provide only the visual description of the content assistant A vibrant red and white taxi, marked with “SILVER CAB” and a phone number, speeds across a wet city street, its motion captured with a blurred background that emphasizes its velocity. The vehicle, likely a modern SUV, glides over rain-slicked asphalt that reflects its bold colors and the white lines of a crosswalk beneath it. Lush green trees and urban structures fade into streaks of motion behind the car, while a distant fountain and a tall, patterned lamppost are barely discernible through the blur. The scene conveys a dynamic, rainy urban atmosphere, with the taxi’s crisp livery standing out against the gray, reflective pavement. user Describe the image in detail in 4-5 sentences. Rules: - Start your response immediately with the description - Do not echo these instructions - Provide only the visual description of the content assistant A vibrant red and white taxi, marked with “SILVER CAB” and a phone number, speeds across a wet city street, its motion captured with a blurred background that emphasizes its velocity. The vehicle, likely a modern SUV,

				glides over rain-slicked asphalt that reflects its bold colors and the white lines of a crosswalk beneath it. Lush green trees and urban structures fade into streaks of motion behind the car, while a distant fountain and a tall, patterned lamppost are barely discernible through the blur. The scene conveys a dynamic, rainy urban atmosphere, with the taxi’s crisp livery standing out against the gray, reflective pavement. car_sample.jpg /content/radiant/data/corpus/car_sample.jpg
2	11.285	car_sample.jpg	1	[VISION_CAPTION] user Describe the image in detail in 4-5 sentences. Rules: - Start your response immediately with the description - Do not echo these instructions - Provide only the visual description of the content assistant A vibrant red and white taxi, marked with “SILVER CAB” and a phone number, speeds across a wet city street, its motion captured with a blurred background that emphasizes its velocity. The vehicle, likely a modern SUV, glides over rain-slicked asphalt that reflects its bold colors and the white lines of a crosswalk beneath it. Lush green trees and urban structures fade into streaks of motion behind the car, while a distant fountain and a tall, patterned lamppost are barely discernible through the blur. The scene conveys a dynamic, rainy urban atmosphere, with the taxi’s crisp livery standing out against the gray, reflective pavement. user Describe the image in detail in 4-5 sentences. Rules: - Start your response immediately with the description - Do not echo these instructions - Provide only the visual description of the content assistant A vibrant red and white taxi, marked with “SILVER CAB” and a phone number, speeds across a wet city street, its motion captured with a blurred background that emphasizes its velocity. The vehicle, likely a modern SUV, glides over rain-slicked asphalt that reflects its bold colors and the white lines of a

				crosswalk beneath it. Lush green trees and urban structures fade into streaks of motion behind the car, while a distant fountain and a tall, patterned lamppost are barely discernible through the blur. The scene conveys a dynamic, rainy urban atmosphere, with the taxi’s crisp livery standing out against the gray, reflective pavement. car_sample.jpg /content/radiant/data/corpus/car_sample.jpg
3	4.712	la_times.pdf	1	$3.66 DESIGNATED AREAS HIGHER © 2025 THURSDAY, SEPTEMBER 25, 2025 latimes.com SHOOTER KILLS ICE DETAINEE, INJURES 2 IN DALLAS Suspect took his own life, authorities say. la_times.pdf /content/radiant/data/corpus/la_times.pdf
4	4.712	la_times.pdf	1	Officials decry threats against federal agents. By Jamie Stengle and Jack Brook DALLAS — A shooter with a rifle opened fire from a nearby roof onto a U.S. la_times.pdf /content/radiant/data/corpus/la_times.pdf
5	4.712	la_times.pdf	1	Im- migration and Customs En- forcement location in Dallas on Wednesday, killing one detainee and wounding two others in a transport van be- fore taking his own life, au- thorities said. la_times.pdf /content/radiant/data/corpus/la_times.pdf
6	4.712	la_times.pdf	1	The suspect has been identified by a law enforce- ment official as 29-year-old Joshua Jahn. The official could not publicly disclose details of the investigation and spoke to the Associated Press on condition of ano- nymity. la_times.pdf /content/radiant/data/corpus/la_times.pdf
7	4.712	la_times.pdf	1	The exact motivation for the attack was not immedi- ately known. FBI Director Kash Patel posted a photo on social media showing a bullet found at the scene with the words “ANTI-ICE” written on it in what ap- peared to be marker. la_times.pdf /content/radiant/data/corpus/la_times.pdf
8	4.712	la_times.pdf	1	Julio Cortez Associated Press Aric Becker

					AFP/Getty Images LAW enforcement agents, top, investigate an apartment building's roof near the shooting on Wednesday in Dallas. la_times.pdf /content/radiant/data/corpus/la_times.pdf
	9	4.712	la_times.pdf	1	Above, more personnel respond to the scene. Trump's assault on free speech splits the right After Jimmy Kimmel suspension, president's allies worry moves could backfire. By Ana Ceballos and Kevin Rector The return of Jimmy Kimmel to ABC's airwaves flipped the political script, for a time aligning the late- night comedian with several conservative figures who staunchly disagree with fed- eral regulators trying to shut him down over free speech — even as President Trump continued to threaten the network. "I want to thank the peo- ple who don't support my show and what I believe, but support my right to share those beliefs anyway," Kim- mel told viewers during his opening monologue Tues- day night. Trump in recent days has ramped up efforts to stifle his political opposition and what he perceives to be libe- ral bias in media coverage through lawsuits and regu- latory actions, a move that has increasingly concerned the president's supporters and influential conservative personalities. about how the "MAGA gang" was trying to score po- litical points from Charlie Kirk's slaying. On a conser- vative podcast, Brendan Carr, a Trump loyalist who heads the Federal Commu- nications Commission, ac- cused Kimmel of "the sickest conduct" and suggested there could be regulatory consequences for local tele- vision stations whose pro- gramming did not serve the public interest. took "Jimmy Kimmel Live!" off the air at ABC last week, some high- profile Trump al- lies worried the threat of regulating speech was tak- ing it too far — and that con- servatives could be next if the federal government were to follow through. la_times.pdf

				/content/radiant/data/corpus/la_times.pdf	
10	4.712	la_times.pdf	1	After Disney “If we embrace the FCC stripping licenses from any- one who says something you disagree with, the next Democrat president who gets in the White House will do this and will come after everyone right of center,” Sen. Ted Cruz (R-Texas), a critic of Kimmel’s, said Wednesday on his podcast, “Verdict With Ted Cruz,” re-affirming previous com- ments in which he likened Carr’s threats to mafia-like maneuvers. la_times.pdf /content/radiant/data/corpus/la_times.pdf	

Confidence is the raw retrieval / reranker score (for example, the `Document.score` produced by Haystack’s retriever or cross-encoder). It is not renormalized or rescaled within this table; values are shown exactly as produced by the retrieval stack, matching the behavior of `retrieval_automerging.py`.

Sources

Ref	Doc ID	Title / File	Page	Level
[S1]	2311446c-af96-43f5-a796-c0b5e8bfae38	car_sample.jpg	1	leaf
[S2]	22f59e05-b77f-45e3-876e-7a598bdefbfb	car_sample.jpg	1	parent
[S3]	734ccc02-0763-4e9f-a197-104b64d6f7b7	la_times.pdf	1	leaf
[S4]	7d48b6d0-f0ec-4c56-bf79-6ea173d39851	la_times.pdf	1	leaf
[S5]	b70b6931-186f-492e-8cbb-73aefc44b3de	la_times.pdf	1	leaf
[S6]	f83257e7-0627-4d77-8fa2-3f0030ba4100	la_times.pdf	1	leaf
[S7]	3a4ec159-b11d-40c0-9a35-452f18076cb1	la_times.pdf	1	leaf
[S8]	0b00617e-689e-404a-bf63-2300174a7894	la_times.pdf	1	leaf
[S9]	241ec97c-dc0a-484c-8af5-	la_times.pdf	1	leaf

	887d866dd2a6			
[S10]	fac44cd8-48b8-43b4-aec7-6c6f5c844cbf	la_times.pdf	1	leaf

Agentic RAG – Evaluation Metrics

Category	Metric	Value	Notes
Retrieval	Q1 – Documents with snippets	10	
Retrieval	Q1 – Total snippets (context)	10	
Retrieval	Q1 – Average snippet confidence (raw score)	6.0263	Uses raw retrieval/reranker score, mirroring retrieval_automerging.py.
Retrieval	Q1 – Max snippet confidence (raw score)	11.2851	Uses raw retrieval/reranker score, mirroring retrieval_automerging.py.
Citations	Q1 – Number of citations	0	
Citations	Q1 – Unique cited documents	0	
Answer	Q1 – Answer length (chars)	95	
Answer	Q1 – Answer length (words)	12	

n in the metric description (e.g. `avg elapsed (n=4)`) refers to the number of telemetry events (agent invocations) used when computing the average.

No telemetry events were recorded.

Retrieval Cache Summary

Enabled	Yes
Backend	QueryCache

Capacity (entries)	256

```
# RAG with multi-turn conversation (i.e. history)
%%time
from agentic_rag_report import run_smoke_test_entry
from IPython.display import HTML

# Agentic RAG (default mode="rag")
html, path = run_smoke_test_entry(
    "config.fast.yaml",
    queries=[
        "What are the dogs playing at the table?",
        "Describe the scene?",
        "What written on the bottle?"
    ],
    mode="rag",
)
HTML(html)
```

```
config.json: 100%                                          791/791 [00:00<00:00, 98.7kB/s]
model.safetensors: 100%                                    133M/133M [00:01<00:00, 100MB/s]
tokenizer_config.json:     1.33k/? [00:00<00:00, 161kB/s]
vocab.txt:     232k/? [00:00<00:00, 16.2MB/s]
tokenizer.json:     711k/? [00:00<00:00, 60.9MB/s]
special_tokens_map.json: 100%                              132/132 [00:00<00:00, 18.0kB/s]
README.md:     3.68k/? [00:00<00:00, 452kB/s]
 WARNING:agents.retriever:Failed to cache BM25 store for 'leaves_chroma': cannot pickle '_queue.SimpleQueue' object
/root/.cache/chroma/onnx_models/all-MiniLM-L6-v2/onnx.tar.gz: 100%|██████████| 79.3M/79.3M [00:08<00:00, 10.2MiB/s]
   CPU times: user 32.3 s, sys: 4.37 s, total: 36.7 s
Wall time: 53.7 s
```

Agentic RAG Report

Configuration

General

Config path	config.fast.yaml
Num queries (report)	3
Index vector store path	data/database/chroma_leaf_store
Index collection name	leaves

Retriever Settings

retrieval.leaf_only	False
retrieval.enable_hybrid	True
retrieval.leaf_top_k	40
retrieval.bm25_top_k	40
retrieval.enable_rerank	True

retrieval.rerank_top_k	32
qe.enabled	True
qe.num_variants	5
prf.enabled	True
prf.docs	8
prf.terms	6

Query 1

Query

```
What are the dogs playing at the table?
```

Answer

The dogs are playing a game of cards, specifically poker.

Top Documents (by retrieval score)

This table groups context snippets by `doc_id` and shows, for each document, the highest-score snippet used as evidence. Scores are normalized to approximately [0.0, 1.0] based on the ceiling of the maximum raw score.

Rank	Doc ID	Title / File	Score	Pag
1	bb3dc791-d337-48db-a1b8-8a291e20baf3	dogs_playing_poker.png	0.993	1

		2	32ced8f6-a9e1-4220-8122-1368862a1d91	stars_and_stripes_2025-05-12.pdf	0.000		23

		3	d10b4949-1776-4e8f-96e9-6a601370af8f	stars_and_stripes_2025-05-12.pdf	0.000		19

4	fd006a50-137a-4c29-ab5c-7ea33b4892ef	whitepaper_embeddings_and_vectorstores.pdf	0.000	9
5	943c2f99-267a-4bc3-b7fd-14cdbff26ff2	whitepaper_Agents.pdf	0.000	6
6	bdc567d8-906a-4594-8a08-9d6a6eebc96a	Solving Domain-Specific problems using LLMs.pdf	0.000	6
7	dfa6632e-c2bb-4f84-9a2a-0b1ad53d72d8	Agent Quality.pdf	0.000	6
8	f9abad0d-05a5-47d5-908d-bb0fa616da87	Prototype to Production.pdf	0.000	6
9	3f579434-e3bd-4704-bd85-2b7694178c1b	Agents.pdf	0.000	5
10	ac178c7e-2af0-4809-8c38-f3d2a8dd5a4d	whitepaper_Agents.pdf	0.000	5

Query 2

Query

```
Describe the scene?
```

Answer

Four anthropomorphic Saint Bernard dogs are gathered around a green felt poker table in a dimly lit, opulent room, engaged in a game of cards. A single turquoise pendant lamp casts a focused glow over the scene, illuminating the dogs' expressive faces, one of whom smokes a pipe while another holds a hand of cards. The table is cluttered with playing cards, poker chips, cigars, and a bottle of Old Saratoga Whiskey on a silver tray. Each dog is seated in a plush, ornate chair, and the background features a framed painting and a dark wooden bookshelf, adding to the atmosphere of a sophisticated, albeit canine, gambling den.

Top Documents (by retrieval score)

This table groups context snippets by `doc_id` and shows, for each document, the highest-score snippet used as evidence. Scores are normalized to approximately [0.0, 1.0] based on the ceiling of the maximum raw score.

Rank	Doc ID	Title / File	Score	Pag
1	bb3dc791-d337-48db-a1b8-8a291e20baf3	dogs_playing_poker.png	0.999	1

2	5d1538e4-a46f-4d16-87ad-d449b0ebc348	whitepaper_Prompt Engineering_v4.pdf	0.003	36
3	f9abad0d-05a5-47d5-908d-bb0fa616da87	Prototype to Production.pdf	0.002	6
4	27504a76-373e-40d2-8b65-bcc62f27d38c	Agent Quality.pdf	0.002	22
5	324c9859-e7da-4e8c-9760-92ef84e69768	whitepaper_Foundational Large Language models & text generation_v2.pdf	0.002	22

6	8671c3c6-7ab0-483f-bffa-473a07eef075	Agent Quality.pdf	0.001	9
7	fd006a50-137a-4c29-ab5c-7ea33b4892ef	whitepaper_embeddings_and_vectorstores.pdf	0.001	9
8	cb80c290-9691-4bd8-98ec-e700eb694536	Prototype to Production.pdf	0.001	9
9	131ff691-9340-4b8e-80bb-f32127fc3ea3	whitepaper_Prompt Engineering_v4.pdf	0.001	9
10	d2d27b21-606c-4a54-8515-d9464b1a251a	whitepaper_Foundational Large Language models & text generation_v2.pdf	0.001	9

Query 3

Query

```
What written on the bottle?
```

Answer

Old Saratoga Whiskey

Top Documents (by retrieval score)

This table groups context snippets by `doc_id` and shows, for each document, the highest-score snippet used as evidence. Scores are normalized to approximately [0.0, 1.0] based on the ceiling of the maximum raw score.

Rank	Doc ID	Title / File	Score	Page	Snippet

1	bb3dc791-d337-48db-a1b8-8a291e20baf3	dogs_playing_poker.png	0.999	1	Four anthropor dogs are gathe poker table in a engaged in a g turquoise pend glow over the s dogs' expressiv smokes a pipe of cards. The ta playing cards, p bottle of Old Sa tray. Each dog i chair, and the b framed painting bookshelf, addi sophisticated, a den. Four anthr dogs are gathe poker table in a engaged in a g turquoise pend glow over the s dogs' expressiv smokes a pipe of cards. The ta playing cards, p bottle of Old Sa tray. Each dog i chair, and the b framed painting bookshelf, addi sophisticated, a den.
2	1f280c5b-28a3-4177-9006-d11f93314849	la_times.pdf	0.000	1	The exact motiv not immedi- at Kash Patel post media showing scene with the on it in what ap la_times.pdf /content/radiar
3	7036ac82-0f23-4d6f-	Context Engineering_ Sessions & Memory.pdf	0.000	23	The agent only turns of the cor

	a58c-f426abd7797b				window) and c 23
4	36ee69c4-6a30-4ced-86fa-dd3806750009	Agent Quality.pdf	0.000	32	They are the to from tasting th the entire culin
5	d10b4949-1776-4e8f-96e9-6a601370af8f	stars_and_stripes_2025-05-12.pdf	0.000	19	Poston Sam Ste Daniel Berger C Schauffele Byeo Fitzpatrick J.J. S Woodland Thor Hojgaard 64-65 196-14 61-67-7 199-11 65-72-6 -8 65-70-67—2 66-67-69—202 67-65—203 -7 68—203 -7 68- 203 -7 63-70-7 -6 65-71-68—2 66-68-70—204 68-71—204 -6 69—204 -6 66- 205 -5 63-70-7 -5 67-71-68—2 67-70-69—206 66-72—206 -4 71—207 -3 GO Thompson Bria Ryan Gerard Be Rodgers Adam Christiaan Bezu Min Woo Lee R McCarthy Mave Novak Will Zala 65-72-70—207 69-70—208 -2 72—208 -2 65- 208 -2 68-70-7 -2 67-71-71—2 69-68-72—209 73-74—209 -1 72—209 -1 66- Americas Open N.J. Purse: $3M

					N.J. Purse: $3M Third Round Je Nelly Korda An Carlota Cigand Lopez Ramirez Lee Elizabeth S Stephanie Kyria Ruoning Yin Ar 202-14 66-71-6 204-12 68-68-6 205-11 67-70-6 -8 67-70-71—2 69-70-70—209 71-69—209 -7 73—210 -6 70- 210 -6 68-71-7 (7), Australia, d
6	8449d453-7928-4e87-b99a-cc864e58397e	Agents_Companion_v2.pdf	0.000	55	They're known and homemade around noon, b moderate occu to set this as yc you there?
7	49ead540-3746-4047-8451-9f462de95a46	Agents_Companion_v2.pdf	0.000	55	They're known and homemade around noon, b moderate occu to set this as yc you there?
8	5acd3234-4048-40b8-b058-e669ea860ca2	whitepaper_Foundational Large Language models & text generation_v2.pdf	0.000	83	Advances in Ne Processing Syst al., 2023, Palm preprint arXiv:2
9	ebbdc272-c816-4bf7-a2a8-452dd1c37f3a	stars_and_stripes_2025-05-12.pdf	0.000	17	ACROSS 1 Croc Golden State sc Paul 13 Broad s 15 — of Sandw 18 Shoe stats 2 battery 23 Badl IRS 28 French c Preparedness k Pyle's org.
10	623acf50-	Solving Domain-Specific	0.000	28	Figure 6 shows

	93a1-4a51-ade2-c77d96816127	problems using LLMs.pdf			applied to an e PaLM 2. Figure review of Med-

Agentic RAG – Evaluation Metrics

Category	Metric	Value	Notes
Retrieval	Q1 – Documents with snippets	24	
Retrieval	Q1 – Total snippets (context)	32	
Retrieval	Q1 – Average snippet confidence (raw score)	0.0932	Uses raw retrieval/reranker score, mirroring retrieval_automerging.py.
Retrieval	Q1 – Max snippet confidence (raw score)	0.9931	Uses raw retrieval/reranker score, mirroring retrieval_automerging.py.
Citations	Q1 – Number of citations	1	
Citations	Q1 – Unique cited documents	0	
Answer	Q1 – Answer length (chars)	57	
Answer	Q1 – Answer length (words)	10	
Retrieval	Q2 – Documents with snippets	30	
Retrieval	Q2 – Total snippets (context)	32	
Retrieval	Q2 – Average snippet confidence (raw score)	0.0950	Uses raw retrieval/reranker score, mirroring retrieval_automerging.py.
Retrieval	Q2 – Max snippet confidence (raw score)	0.9993	Uses raw retrieval/reranker score, mirroring retrieval_automerging.py.
Citations	Q2 – Number of citations	1	
Citations	Q2 – Unique cited	0	

Citations	Q2 – Unique cited documents	0	
Answer	Q2 – Answer length (chars)	628	
Answer	Q2 – Answer length (words)	108	
Retrieval	Q3 – Documents with snippets	23	
Retrieval	Q3 – Total snippets (context)	32	
Retrieval	Q3 – Average snippet confidence (raw score)	0.0937	Uses raw retrieval/reranker score, mirroring retrieval_automerging.py.
Retrieval	Q3 – Max snippet confidence (raw score)	0.9987	Uses raw retrieval/reranker score, mirroring retrieval_automerging.py.
Citations	Q3 – Number of citations	1	
Citations	Q3 – Unique cited documents	0	
Answer	Q3 – Answer length (chars)	20	
Answer	Q3 – Answer length (words)	3	
Agents	BasicCriticAgent – avg elapsed (n=4)	0.03 ms	Average elapsed wall time per telemetry event.
Agents	BasicDecompositionAgent – avg elapsed (n=3)	0.03 ms	Average elapsed wall time per telemetry event.
Agents	BasicGuardrailAgent – avg elapsed (n=3)	0.07 ms	Average elapsed wall time per telemetry event.
Agents	BasicPRFAgent – avg elapsed (n=3)	39.05 ms	Average elapsed wall time per telemetry event.
Agents	BasicPlannerAgent – avg elapsed (n=3)	0.07 ms	Average elapsed wall time per telemetry event.
Agents	BasicPolicyAgent – avg elapsed (n=4)	0.03 ms	Average elapsed wall time per telemetry event.
Agents	BasicPostProcessorAgent – avg elapsed (n=3)	0.04 ms	Average elapsed wall time per telemetry event.
Agents	BasicRerankAgent – avg elapsed (n=3)	463.47 ms	Average elapsed wall time per telemetry event.

	elapsed (n=3)	ms	telemetry event.
Agents	BasicRouterAgent – avg elapsed (n=3)	0.07 ms	Average elapsed wall time per telemetry event.
Agents	HybridRetrievalAgent – avg elapsed (n=3)	9646.03 ms	Average elapsed wall time per telemetry event.
Agents	LLMGeneratorAgent – avg elapsed (n=4)	1745.76 ms	Average elapsed wall time per telemetry event.
Agents	LLMQEAgent – avg elapsed (n=2)	2360.86 ms	Average elapsed wall time per telemetry event.
Agents	LLMQueryRewriteAgent – avg elapsed (n=2)	1763.28 ms	Average elapsed wall time per telemetry event.

n in the metric description (e.g. `avg elapsed (n=4)`) refers to the number of telemetry events (agent invocations) used when computing the average.

Telemetry

Agent	Event	Phase	Elapsed	Model
BasicRouterAgent	router.decided	QUERY	0.07 ms	gemma3 cloud
BasicDecompositionAgent	decomposition.done	QUERY	0.04 ms	gemma3 cloud

BasicPlannerAgent	planner.plan	QUERY	0.07 ms	gemma3 cloud
BasicGuardrailAgent	guardrail.validate_plan	QUERY	0.06 ms	gemma3 cloud
BasicPRFAgent	prf.compute	QUERY	28.86 ms	gemma3 cloud
LLMQEAgent	qe.expand	QUERY	2872.84 ms	gemma3 cloud
HybridRetrievalAgent	retriever.results	QUERY	18596.37 ms	gemma3 cloud
BasicRerankAgent	rerank.rerank	QUERY	711.92 ms	gemma3 cloud

LLMGeneratorAgent	generator.output	QUERY	1633.36 ms	gemma3 cloud
BasicCriticAgent	critic.evaluate	QUERY	0.03 ms	gemma3 cloud
BasicPolicyAgent	policy.decision	QUERY	0.04 ms	gemma3 cloud
LLMGeneratorAgent	generator.output	QUERY	1231.28 ms	gemma3 cloud
BasicCriticAgent	critic.evaluate	QUERY	0.02 ms	gemma3 cloud
BasicPolicyAgent	policy.decision	QUERY	0.03 ms	gemma3 cloud
BasicPostProcessorAgent	postprocess.format	QUERY	0.04 ms	gemma3 cloud
BasicRouterAgent	router.decided	QUERY	0.06 ms	gemma3

	[illegible]	[illegible]	[illegible]	[illegible]	[illegible] cloud
	BasicDecompositionAgent	decomposition.done	QUERY	0.03 ms	gemma3 cloud
	BasicPlannerAgent	planner.plan	QUERY	0.06 ms	gemma3 cloud
	BasicGuardrailAgent	guardrail.validate_plan	QUERY	0.05 ms	gemma3 cloud

	LLMQueryRewriteAgent	query.rewritten_with_history	QUERY	2096.67 ms	gemma3 cloud
	BasicPRFAgent	prf.compute	QUERY	36.69 ms	gemma3 cloud
	LLMQEAgent	qe.expand	QUERY	1848.89 ms	gemma3 cloud
	HybridRetrievalAgent	retriever.results	QUERY	8497.95 ms	gemma3 cloud
	BasicRerankAgent	rerank.rerank	QUERY	452.78 ms	gemma3 cloud
	LLMGeneratorAgent	generator.output	QUERY	2839.71 ms	gemma3 cloud
	BasicCriticAgent	critic.evaluate	QUERY	0.04 ms	gemma3 cloud

	BasicPolicyAgent	policy.decision	QUERY	0.03 ms	gemma3 cloud
	BasicPostProcessorAgent	postprocess.format	QUERY	0.04 ms	gemma3 cloud
	BasicRouterAgent	router.decided	QUERY	0.07 ms	gemma3 cloud
	BasicDecompositionAgent	decomposition.done	QUERY	0.03 ms	gemma3 cloud
	BasicPlannerAgent	planner.plan	QUERY	0.07 ms	gemma3 cloud

	BasicGuardrailAgent	guardrail.validate_plan	QUERY	0.09 ms	gemma3 cloud
	LLMQueryRewriteAgent	query.rewritten_with_history	QUERY	1429.88 ms	gemma3 cloud
	BasicPRFAgent	prf.compute	QUERY	51.59 ms	gemma3 cloud
	HybridRetrievalAgent	retriever.results	QUERY	1843.78 ms	gemma3 cloud
	BasicRerankAgent	rerank.rerank	QUERY	225.71 ms	gemma3 cloud

LLMGeneratorAgent	generator.output	QUERY	1278.67 ms	gemma3 cloud
BasicCriticAgent	critic.evaluate	QUERY	0.02 ms	gemma3 cloud
BasicPolicyAgent	policy.decision	QUERY	0.02 ms	gemma3 cloud
BasicPostProcessorAgent	postprocess.format	QUERY	0.03 ms	gemma3 cloud

Phases
`WARMUP`: non-user warmup calls.
`QUERY`: normal user query handling.
`INDEX`: indexing / corpus processing.
`SYSTEM`: diagnostics or background work.

Retrieval Cache Summary

Enabled	Yes
Backend	QueryCache
Capacity (entries)	256
Current size (entries)	3
Hits	0
Misses	3
Stores	3

Total lookups	3
Hit rate	0.0%

Hit rate is the percentage of retrieval requests served from the cache instead of hitting the vector store / retriever.

```
# RAG only
%%time
from agentic_rag_report import run_smoke_test_entry
from IPython.display import HTML

# or provide your own list of queries
html, path = run_smoke_test_entry(
    "config.fast.yaml",
    queries=[
        "Is there an image of a taxi in the documents?"
    ],
)
HTML(html)
```

```
CPU times: user 11.8 s, sys: 1.33 s, total: 13.1 s
Wall time: 13.5 s
```

Agentic RAG Report

Configuration

General

Config path	config.fast.yaml
Num queries (report)	1
Index vector store path	data/database/chroma_leaf_store
Index collection name	leaves

Retriever Settings

retrieval.leaf_only	False
retrieval.enable_hybrid	True
retrieval.leaf_top_k	40
retrieval.bm25_top_k	40
retrieval.enable_rerank	True
retrieval.rerank_top_k	32
qe.enabled	True
qe.num_variants	5
prf.enabled	True
prf.docs	8
prf.terms	6

Query 1

Query

```
Is there an image of a taxi in the documents?
```

Answer

Yes, there is an image of a vibrant red and white taxi in the documents.

Top Documents (by retrieval score)

This table groups context snippets by `doc_id` and shows, for each document, the highest-score snippet used as evidence. Scores are normalized to approximately [0.0, 1.0] based on the ceiling of the maximum raw score.

Rank	Doc ID	Title / File	Score	Pag
1	22f59e05-b77f-45e3-876e-7a598bdefbfb	car_sample.jpg	0.856	1
2	9575b1bc-f4f7-4314-a5db-a93ab573f0ed	whitepaper_embeddings_and_vectorstores.pdf	0.001	24

3	e7a79b34-a7f2-48c0-9d57-76a4969ddaa0	ocr_us_constitution.png	0.000	1
4	cc5e8666-f25e-4bf0-b424-4e67c1b7ba1e	ocr_us_constitution.png	0.000	1

5	5d1538e4-a46f-4d16-87ad-d449b0ebc348	whitepaper_Prompt Engineering_v4.pdf	0.000	36
6	0462c49d-1873-4f35-924a-47d23651efb6	whitepaper_Foundational Large Language models & text generation_v2.pdf	0.000	34
7	6b37d880-cebd-49cb-8458-090bf2fce356	whitepaper_embeddings_and_vectorstores.pdf	0.000	34
8	4c8c5671-a2f8-47f0-b624-560c8d3679dd	Agents_Companion_v2.pdf	0.000	27

9	000f147b-c606-43ce-80c0-5bb118b54061	ocr_us_constitution.png	0.000	1
10	91de21a2-f97f-46e1-bd70-734bf4b333cd	Context Engineering_ Sessions & Memory.pdf	0.000	9

Agentic RAG – Evaluation Metrics

Category	Metric	Value	Notes
Retrieval	Q1 – Documents with snippets	17	
Retrieval	Q1 – Total snippets (context)	32	
Retrieval	Q1 – Average snippet confidence (raw score)	0.0536	Uses raw retrieval/reranker score, mirroring retrieval_automerging.py.
Retrieval	Q1 – Max snippet confidence (raw score)	0.8558	Uses raw retrieval/reranker score, mirroring retrieval_automerging.py.
Citations	Q1 – Number of citations	1	
Citations	Q1 – Unique cited documents	0	
Answer	Q1 – Answer length (chars)	72	
Answer	Q1 – Answer length (words)	15	

Agents	BasicCriticAgent – avg elapsed (n=2)	0.02 ms	Average elapsed wall time per telemetry event.
Agents	BasicDecompositionAgent – avg elapsed (n=1)	0.02 ms	Average elapsed wall time per telemetry event.
Agents	BasicGuardrailAgent – avg elapsed (n=1)	0.05 ms	Average elapsed wall time per telemetry event.
Agents	BasicPRFAgent – avg elapsed (n=1)	30.51 ms	Average elapsed wall time per telemetry event.
Agents	BasicPlannerAgent – avg elapsed (n=1)	0.05 ms	Average elapsed wall time per telemetry event.
Agents	BasicPolicyAgent – avg elapsed (n=2)	0.02 ms	Average elapsed wall time per telemetry event.
Agents	BasicPostProcessorAgent – avg elapsed (n=1)	0.03 ms	Average elapsed wall time per telemetry event.
Agents	BasicRerankAgent – avg elapsed (n=1)	376.89 ms	Average elapsed wall time per telemetry event.
Agents	BasicRouterAgent – avg elapsed (n=1)	0.05 ms	Average elapsed wall time per telemetry event.
Agents	HybridRetrievalAgent – avg elapsed (n=1)	8150.91 ms	Average elapsed wall time per telemetry event.
Agents	LLMGeneratorAgent – avg elapsed (n=2)	1573.64 ms	Average elapsed wall time per telemetry event.
Agents	LLMQEAgent – avg elapsed (n=1)	1808.80 ms	Average elapsed wall time per telemetry event.

n in the metric description (e.g. `avg elapsed (n=4)`) refers to the number of telemetry events (agent invocations) used when computing the average.

Telemetry

Agent	Event	Phase	Elapsed	Model
BasicRouterAgent	router.decided	QUERY	0.05 ms	gemma3:12b-cloud

	BasicDecompositionAgent	decomposition.done	QUERY	0.02 ms	gemma3:12b-cloud
	BasicPlannerAgent	planner.plan	QUERY	0.05 ms	gemma3:12b-cloud
	BasicGuardrailAgent	guardrail.validate_plan	QUERY	0.05 ms	gemma3:12b-cloud
	BasicPRFAgent	prf.compute	QUERY	30.51	gemma3:12b-

[illegible]	[illegible]	[illegible]	ms	cloud
LLMQEAgent	qe.expand	QUERY	1808.80 ms	gemma3:12b-cloud
HybridRetrievalAgent	retriever.results	QUERY	8150.91 ms	gemma3:12b-cloud
BasicRerankAgent	rerank.rerank	QUERY	376.89 ms	gemma3:12b-cloud
LLMGeneratorAgent	generator.output	QUERY	1625.31 ms	gemma3:12b-cloud
BasicCriticAgent	critic.evaluate	QUERY	0.02 ms	gemma3:12b-cloud
BasicPolicyAgent	policy.decision	QUERY	0.03 ms	gemma3:12b-cloud
LLMGeneratorAgent	generator.output	QUERY	1521.97 ms	gemma3:12b-cloud
BasicCriticAgent	critic.evaluate	QUERY	0.02 ms	gemma3:12b-cloud

BasicPolicyAgent	policy.decision	QUERY	0.02 ms	gemma3:12b-cloud
BasicPostProcessorAgent	postprocess.format	QUERY	0.03 ms	gemma3:12b-cloud

Phases
`WARMUP`: non-user warmup calls.
`QUERY`: normal user query handling.
`INDEX`: indexing / corpus processing.
`SYSTEM`: diagnostics or background work.

Retrieval Cache Summary

Enabled	Yes
Backend	QueryCache
Capacity (entries)	256
Current size (entries)	5
Hits	0
Misses	5
Stores	5
Total lookups	5
Hit rate	0.0%

Hit rate is the percentage of retrieval requests served from the cache instead of hitting the vector store / retriever.

```
# RAG with multi-turn conversation (i.e. history)
# Note this is a case where history should be disabled in config since ques
# are not related for cleaner LLM generation (i.e. dont mix incorrect histo
# with unrelated query). Set include_in_prompt to false.
%%time
from agentic_rag_report import run_smoke_test_entry
from IPython.display import HTML

# or provide your own list of queries
html, path = run_smoke_test_entry(
    "config.fast.yaml",
    queries=[
        "What does Article I outline in the Constitution?",
        "What is A2A and how is it used?",
        "What did Zelensky comment about Putin and Russia's war with Ukrair
    ],
)
HTML(html)
```

```
CPU times: user 28.3 s, sys: 3.5 s, total: 31.8 s
Wall time: 45.5 s
```

Agentic RAG Report

Configuration

General

Config path	config.fast.yaml
Num queries (report)	3
Index vector store path	data/database/chroma_leaf_store
Index collection name	leaves

Retriever Settings

retrieval.leaf_only	False
retrieval.enable_hybrid	True
retrieval.leaf_top_k	40
retrieval.bm25_top_k	40
retrieval.enable_rerank	True
retrieval.rerank_top_k	32
qe.enabled	True
qe.num_variants	5
prf.enabled	True
prf.docs	8
prf.terms	6

Query 1

Query

```
What does Article I outline in the Constitution?
```

Answer

According to the provided text, Article I outlines the structure of the legislative branch. It details the composition and powers of Congress, including the House of Representatives and the Senate, as well as rules for elections, representation, and legislative procedures.

Top Documents (by retrieval score)

This table groups context snippets by `doc_id` and shows, for each document, the highest-score snippet used as evidence. Scores are normalized to approximately [0.0, 1.0] based on the ceiling of the maximum raw score.

Rank	Doc ID	Title / File	Score	Pag
1	e7a79b34-a7f2-48c0-9d57-76a4969ddaa0	ocr_us_constitution.png	0.991	1

		2	88700799-3fba-4842-bf2f-432f818ed777	ocr_us_constitution.png	0.001		1

3	c81ca6fd-2f95-4540-9fc6-c51ed9f4822f	Prototype to Production.pdf	0.000		17
4	cc5e8666-f25e-4bf0-b424-4e67c1b7ba1e	ocr_us_constitution.png	0.000		1

5	5d594d07-1d4a-4c98-8bfa-c911c17b07a2	Introduction to Agents.pdf	0.000		23
6	f69e0c78-b7b7-47fb-9d37-d1dcc8d0a8b4	la_times.pdf	0.000		1

7	bfc92e00-ac3b-4632-b687-fbd55e5ba073	stars_and_stripes_2025-05-12.pdf	0.000	3

		8	f71f2c5a-5b21-4530-bb0d	whitepaper_Foundational Large Language models & text generation_v2.pdf	0.000		84

	bb0d- 17d732572d2b				
9	d67e5634- 7591-4429- 8eb6- a9f7ea952ee5	whitepaper_embeddings_and_vectorstores.pdf	0.000		62
10	42460a71- f87d-4f2b- aa3f- 76e6a5abd402	Agent Tools & Interoperability with Model Context Protocol (MCP).pdf	0.000		53

Query 2

Query

```
What is A2A and how is it used?
```

Answer

The provided text does not mention Article I of the Constitution. However, according to the text, the Agent2Agent (A2A) protocol is an open standard designed to solve the problem of connecting different specialized agents built by different teams within an enterprise. It acts as a "universal handshake for the agentic economy," allowing agents to discover each other and communicate.

Here's how it's used, according to the text:

*   **Discovery:** A2A allows agents to publish a digital "business card" (Agent Card) advertising their capabilities and how to interact with them.
*   **Communication:** It enables agents to communicate using a task-oriented architecture, framing interactions as asynchronous "tasks."
*   **Collaboration:** It facilitates seamless collaboration between agents, as seen in scenarios like a fraud detection agent collaborating with a transaction analysis agent.
*   **Examples:** The text provides examples of A2A usage, such as a mechanic agent communicating with a parts supplier agent, and a customer using A2A to communicate with a "Shop Manager" agent.

Top Documents (by retrieval score)

This table groups context snippets by `doc_id` and shows, for each document, the highest-score snippet used as evidence. Scores are normalized to approximately [0.0, 1.0] based on the ceiling of the maximum raw score.

Rank	Doc ID	Title / File	Score	Page	Snippet
1	b15d9c2d-b15a-462d-a2fd-cabc9af521df	Introduction to Agents.pdf	0.610	33	A2A transforms a agents into a true ecosystem. 33
2	517b742c-96fe-4b3d-ab84-e18d64e3e6e3	Prototype to Production.pdf	0.606	28	A2A Protocol: Fro Implementation T designed to break silos and enable s between agents. C where a fraud det suspicious activity full context, it nee separate transacti

					Without A2A, a h manually bridge t that could take ho agents collaborate resolving the issue
3	bcf6495d-2463-435b-89ce-243542a36943	Prototype to Production.pdf	0.571	34	Agent-to-Agent (the issue, the Mec determines a part A2A to communic "Parts Supplier" a availability and pl Prototype to Prod /content/radiant/ to Production.pdf
4	e1c82e8b-84aa-4254-a605-db0dd8d4d399	Prototype to Production.pdf	0.227	35	In this workflow, A higher-level, conv oriented interactio customer, the sho external suppliers.
5	cd7b585d-a9f6-4689-8313-b187fb74b1bf	Prototype to Production.pdf	0.141	32	Second is robust s A2A interactions a requiring a sophis layer for tracking transactional integ
6	4f72f979-30b2-4ac5-bcc0-372cc58e4391	Prototype to Production.pdf	0.135	27	A2A - Reusability You've built dozen agents across you customer service t agent.
7	e7a79b34-a7f2-48c0-9d57-76a4969ddaa0	ocr_us_constitution.png	0.125	1	The image display “THE UNITED STAT with a subtitle not LAYOUT BY JOHN RATIFIED CONTEN formatted in an ol resembling histori and is organized i sections, beginnin outlines the struct branch. The page with legal text, de composition and p including the Hou

						and the Senate, as elections, represer procedures. At the indicates "PAGE O VERIFIED WITH TH ORIGINAL, POSTE 2017, CREATIVE C 4.0." The overall a historical reprodu poster, with a forr tone. THE UNITED CONSTITUTION P JOHN CARTER, 17 CONTENT E, the P S TAT E S, in order fect union, establi domestic tranquili common defense, welfare, and secur liberty to ourselve do ordain and est tion for the United
	8	ce020f10-3920-4809-b172-01f46fdcc29a	Context Engineering_ Sessions & Memory.pdf	0.121	20	One emerging arc architectural patte collaboration betv agents is Agent-to communication8. enables agents to it fails to address sharing rich, conte agent's conversati in its framework's result, any A2A m session events rec layer to be useful. architectural patte involves abstractir into a framework- such as Memory. which preserves ra specific objects lik Messsages, a men to hold processed information. Key i summaries, extrac

					summaries, extrac —is extracted fror and is typically stc dictionaries. The r structures are not framework's interi representation, wl as a universal, cor Context Engineeri Memory.pdf /content/radiant/(Engineering_ Sess
9	a9a2548d-738c-4d72-b339-d420ce0b5c6f	Context Engineering_ Sessions & Memory.pdf	0.029	18	# Parent agent. ro ..., sub_agents=[su sub_agent_2]) Sni communication ac frameworks In the histories model, e its own private co functions like a bl agents. All interna intermediary thou reasoning steps— agent's private log to others. Commu through explicit m agent shares its fi process. This inter implemented by e Agent-as-a-tool o Agent (A2A) Proto Tool, one agent in were a standard t and receiving a fir output6. With the (A2A) Protocol, ag protocol for direc explore the A2A p in the next sessior Engineering_ Sess /content/radiant/(Engineering_ Sess
10	fbba0d6b-cf98-4a2b-8b28-	Introduction to Agents.pdf	0.026	53	30. A2UI is a proto via structured out passing

	6bfe56f1d4f1				https://github.cor https://cloud.goog ai/generative- ai/docs/models/g api.

Query 3

Query

```
What did Zelensky comment about Putin and Russia's war with Ukraine?
```

Answer

According to the provided text, Zelensky asserted that Vladimir Putin wants to expand his war in Europe. He also stated that the world is in “the most destructive arms race in human history” and urged the international community to act against Russia now.

Top Documents (by retrieval score)

This table groups context snippets by `doc_id` and shows, for each document, the highest-score snippet used as evidence. Scores are normalized to approximately [0.0, 1.0] based on the ceiling of the maximum raw score.

Rank	Doc ID	Title / File	Score	Page	Snippet
1	e73c533c-d95f-4b64-bf63-53970fe2f2a3	la_times.pdf	0.995	1	A12 UNITED NATI dymyr Zelensky to the world is in “the hu- man history” a community to act Vladimir Putin war bleak view of toda level meeting of th international instit Nations haven’t be Gaza, Sudan and e law can’t help nati survives,” the Ukra curity guarantees Zelensky spoke fro

					chamber a day afte who expressed for Russia. support Tru Ukraine could win dramatic shift from Kyiv to make conc Russian President its smaller neighbo the surprise U.S. pi good meeting" wit "strong leaders." " said, expressing ap United States and member nations to dragging this war Ukraine, A4] la_tim /content/radiant/c
2	f69e0c78-b7b7-47fb-9d37-d1dcc8d0a8b4	la_times.pdf	0.985	1	"Data centers are p Assemblymem- be bill's author. [See V Putin, AI, wider wa la_times.pdf /content/radiant/c
3	060cdcde-cff7-4e56-ba5e-68a592af6965	stars_and_stripes_2025-05-12.pdf	0.973	14	Monday, May 12, 2 overtures Associat President Volodym Russia's of- fer for there must be a fu before negotiation called Russian Pres offer to start talks sign," and said tha waiting for this for Ukraine did not ag Russia unilat- erall repeatedly violatin peared to insist on unconditional ceas point in continuing We expect Russia t lasting, and reliabl and Uk- raine is re added, however, th ending any war is

					resumed mass dro Sunday, after its se Russia launched 1(drones from six dif Force said on Sund down and an- othe reach targets due t Russian Defense M of “vi- olating” Mo than 14,000 times. /content/radiant/c 05-12.pdf
4	57af57d9-cc03-4efb-b0d3-3e1d80dc4126	stars_and_stripes_2025-05-12.pdf	0.665	4	Yang, speaking to noted how North I Russia by covertly am- munition for N Ukraine. In June, K Putin pledged mut at- tacked. “We sp Korea with resourc for Pyongyang’s te Yang said.
5	97420a74-4116-4c49-952b-66aaf5f2c3ca	stars_and_stripes_2025-05-12.pdf	0.174	16	Two years ago, tha crimes charges for tion of Ukrainian c Lvova-Belova, a Ru title is commission
6	00f3be3b-3701-4aaa-bbb8-ecc45ab90a43	stars_and_stripes_2025-05-12.pdf	0.002	16	Still, it’s a mistake eventual courtroor and other war crim measure of accour
7	890ccda7-b4f4-47d0-8894-3c115de25317	stars_and_stripes_2025-05-12.pdf	0.001	16	David Von Drehle i Washington Post a the author of four That Changed Ame a disgusting possib to The Washington stench of Russian v sified as crimes ag Nations-mandatec mountains of re- p
8	57447011-	stars_and_stripes_2025-	0.001	14	PAGE14 • S T A R S

			1fd5-4fa5-9f6e-0d2ab5018305	05-12.pdf			calls for peace BY I DELL'ORTO Associ Leo XIV called for a and an immedi- at lease of hostages a his first Sun- day n address the world' ever-pre- sent call the loggia of St. Pe 100,000 people be had returned to th the world on Thurs election as pope, t Then too he de- liv ARMANGUE/AP A Pope Leo XIV appe Peter's Basilica for given to him by his quoted Pope Franc conflicts ravaging "third world war in suffer- ings of the said. "Let everythir genuine, just and l Leo was picking up Sunday blessing at Whereas his prede from the studio wi the side of the piaz the square and the
		9	b15d9c2d-b15a-462d-a2fd-cabc9af521df	Introduction to Agents.pdf	0.000	33	It makes the agent partner. Agents an connect with huma each other. As an e different teams wil Without a commo require building a integrations that a challenge is twofol find other agents a communication (h same language?). the open standard acts as a universal economy. A2A allo

					"business card," kr JSON file advertise network endpoint, required to interac /content/radiant/c Agents.pdf
10	ce020f10-3920-4809-b172-01f46fdcc29a	Context Engineering_ Sessions & Memory.pdf	0.000	20	One emerging arcl pattern for coordir isolated agents is communication8. to exchange messa problem of sharing agent's conversati framework's intern message containin translation layer to architectural patte abstracting shared agnostic data laye store, which prese objects like Events designed to hold Key information— and facts—is extra typically stored as layer's data structu framework's intern allows it to serve a This pattern allows true collaborative cognitive resource translators. Produc When moving an its session manage simple log to a rob key considerations security and privac A managed sessio is specifically desig requirements. 20

Agentic RAG – Evaluation Metrics

Category	Metric	Value	Notes
Retrieval	Q1 – Documents with snippets	16	
Retrieval	Q1 – Total snippets (context)	32	
Retrieval	Q1 – Average snippet confidence (raw score)	0.1551	Uses raw retrieval/reranker score, mirroring retrieval_automerging.py.
Retrieval	Q1 – Max snippet confidence (raw score)	0.9911	Uses raw retrieval/reranker score, mirroring retrieval_automerging.py.
Citations	Q1 – Number of citations	1	
Citations	Q1 – Unique cited documents	0	
Answer	Q1 – Answer length (chars)	273	
Answer	Q1 – Answer length (words)	40	
Retrieval	Q2 – Documents with snippets	18	
Retrieval	Q2 – Total snippets (context)	32	
Retrieval	Q2 – Average snippet confidence (raw score)	0.1908	Uses raw retrieval/reranker score, mirroring retrieval_automerging.py.
Retrieval	Q2 – Max snippet confidence (raw score)	0.6101	Uses raw retrieval/reranker score, mirroring retrieval_automerging.py.
Citations	Q2 – Number of citations	1	
Citations	Q2 – Unique cited documents	0	
Answer	Q2 – Answer length (chars)	1090	
Answer	Q2 – Answer length (words)	160	
Retrieval	Q3 – Documents with snippets	14	

Retrieval	Q3 – Total snippets (context)	32	
Retrieval	Q3 – Average snippet confidence (raw score)	0.3456	Uses raw retrieval/reranker score, mirroring retrieval_automerging.py.
Retrieval	Q3 – Max snippet confidence (raw score)	0.9953	Uses raw retrieval/reranker score, mirroring retrieval_automerging.py.
Citations	Q3 – Number of citations	1	
Citations	Q3 – Unique cited documents	0	
Answer	Q3 – Answer length (chars)	255	
Answer	Q3 – Answer length (words)	43	
Agents	BasicCriticAgent – avg elapsed (n=6)	0.03 ms	Average elapsed wall time per telemetry event.
Agents	BasicDecompositionAgent – avg elapsed (n=3)	0.04 ms	Average elapsed wall time per telemetry event.
Agents	BasicGuardrailAgent – avg elapsed (n=3)	0.05 ms	Average elapsed wall time per telemetry event.
Agents	BasicPRFAgent – avg elapsed (n=3)	41.19 ms	Average elapsed wall time per telemetry event.
Agents	BasicPlannerAgent – avg elapsed (n=3)	0.05 ms	Average elapsed wall time per telemetry event.
Agents	BasicPolicyAgent – avg elapsed (n=6)	0.03 ms	Average elapsed wall time per telemetry event.
Agents	BasicPostProcessorAgent – avg elapsed (n=3)	0.03 ms	Average elapsed wall time per telemetry event.
Agents	BasicRerankAgent – avg elapsed (n=3)	343.49 ms	Average elapsed wall time per telemetry event.
Agents	BasicRouterAgent – avg elapsed (n=3)	0.05 ms	Average elapsed wall time per telemetry event.
Agents	HybridRetrievalAgent – avg elapsed (n=3)	6174.49 ms	Average elapsed wall time per telemetry event.
Agents	LLMGeneratorAgent – avg elapsed (n=6)	2845.03 ms	Average elapsed wall time per telemetry event.

	elapsed (n=6)	ms	telemetry event.
Agents	LLMQEAgent – avg elapsed (n=3)	1915.73 ms	Average elapsed wall time per telemetry event.
Agents	LLMQueryRewriteAgent – avg elapsed (n=2)	1501.76 ms	Average elapsed wall time per telemetry event.

n in the metric description (e.g. `avg elapsed (n=4)`) refers to the number of telemetry events (agent invocations) used when computing the average.

Telemetry

Agent	Event	Phase	Elapsed	Model
BasicRouterAgent	router.decided	QUERY	0.06 ms	gemma3: cloud
BasicDecompositionAgent	decomposition.done	QUERY	0.02 ms	gemma3: cloud
BasicPlannerAgent	planner.plan	QUERY	0.05 ms	gemma3: cloud

	BasicGuardrailAgent	guardrail.validate_plan	QUERY	0.05 ms	gemma3: cloud
	BasicPRFAgent	prf.compute	QUERY	27.42 ms	gemma3: cloud
	LLMQEAgent	qe.expand	QUERY	1969.96 ms	gemma3: cloud
	HybridRetrievalAgent	retriever.results	QUERY	7053.04 ms	gemma3: cloud
	BasicRerankAgent	rerank.rerank	QUERY	375.01 ms	gemma3: cloud
	LLMGeneratorAgent	generator.output	QUERY	2085.39 ms	gemma3: cloud
	BasicCriticAgent	critic.evaluate	QUERY	0.02 ms	gemma3: cloud

	BasicPolicyAgent	policy.decision	QUERY	0.03 ms	gemma3: cloud
	LLMGeneratorAgent	generator.output	QUERY	1774.33 ms	gemma3: cloud
	BasicCriticAgent	critic.evaluate	QUERY	0.02 ms	gemma3: cloud
	BasicPolicyAgent	policy.decision	QUERY	0.02 ms	gemma3: cloud
	BasicPostProcessorAgent	postprocess.format	QUERY	0.03 ms	gemma3: cloud
	BasicRouterAgent	router.decided	QUERY	0.05 ms	gemma3: cloud

	BasicDecompositionAgent	decomposition.done	QUERY	0.05 ms	gemma3: cloud
	BasicPlannerAgent	planner.plan	QUERY	0.05 ms	gemma3: cloud
	BasicGuardrailAgent	guardrail.validate_plan	QUERY	0.05 ms	gemma3: cloud

	LLMQueryRewriteAgent	query.rewritten_with_history	QUERY	1585.03 ms	gemma3: cloud
	BasicPRFAgent	prf.compute	QUERY	45.01 ms	gemma3: cloud
	LLMQEAgent	qe.expand	QUERY	1960.78 ms	gemma3: cloud
	HybridRetrievalAgent	retriever.results	QUERY	7208.70 ms	gemma3: cloud
	BasicRerankAgent	rerank.rerank	QUERY	406.70 ms	gemma3: cloud
	LLMGeneratorAgent	generator.output	QUERY	4476.81 ms	gemma3: cloud
	BasicCriticAgent	critic.evaluate	QUERY	0.04 ms	gemma3: cloud
	BasicPolicyAgent	policy.decision	QUERY	0.03 ms	gemma3: cloud

LLMGeneratorAgent	generator.output	QUERY	4653.38 ms	gemma3: cloud
BasicCriticAgent	critic.evaluate	QUERY	0.03 ms	gemma3: cloud
BasicPolicyAgent	policy.decision	QUERY	0.02 ms	gemma3: cloud
BasicPostProcessorAgent	postprocess.format	QUERY	0.03 ms	gemma3: cloud
BasicRouterAgent	router.decided	QUERY	0.04 ms	gemma3: cloud
BasicDecompositionAgent	decomposition.done	QUERY	0.03 ms	gemma3:

	BasicDecompositionAgent	decomposition.done	QUERY	0.05 ms	gemma3: cloud
	BasicPlannerAgent	planner.plan	QUERY	0.05 ms	gemma3: cloud
	BasicGuardrailAgent	guardrail.validate_plan	QUERY	0.05 ms	gemma3: cloud
	LLMQueryRewriteAgent	query.rewritten_with_history	QUERY	1418.49 ms	gemma3: cloud

	BasicPRFAgent	prf.compute	QUERY	51.14 ms	gemma3: cloud
	LLMQEAgent	qe.expand	QUERY	1816.44 ms	gemma3: cloud
	HybridRetrievalAgent	retriever.results	QUERY	4261.73 ms	gemma3: cloud
	BasicRerankAgent	rerank.rerank	QUERY	248.76 ms	gemma3: cloud
	LLMGeneratorAgent	generator.output	QUERY	1968.73 ms	gemma3: cloud
	BasicCriticAgent	critic.evaluate	QUERY	0.02 ms	gemma3: cloud
	BasicPolicyAgent	policy.decision	QUERY	0.03 ms	gemma3: cloud

LLMGeneratorAgent	generator.output	QUERY	2111.51 ms	gemma3: cloud
BasicCriticAgent	critic.evaluate	QUERY	0.02 ms	gemma3: cloud
BasicPolicyAgent	policy.decision	QUERY	0.04 ms	gemma3: cloud
BasicPostProcessorAgent	postprocess.format	QUERY	0.03 ms	gemma3: cloud

Phases
`WARMUP`: non-user warmup calls.
`QUERY`: normal user query handling.
`INDEX`: indexing / corpus processing.
`SYSTEM`: diagnostics or background work.

Retrieval Cache Summary

Enabled	Yes
Backend	QueryCache
Capacity (entries)	256
Current size (entries)	8
Hits	0
Misses	8

Stores	8
Total lookups	8
Hit rate	0.0%

Hit rate is the percentage of retrieval requests served from the cache instead of hitting the vector store / retriever.

```
!python agentic_rag_report.py --config config.fast.yaml --query "Elaborate
```

```
2025-12-14 22:19:43.704474: E external/local_xla/xla/stream_executor/cuda/cuda_f
WARNING: All log messages before absl::InitializeLog() is called are written to
E0000 00:00:1765750783.726015   32591 cuda_dnn.cc:8579] Unable to register cuDNN
E0000 00:00:1765750783.732763   32591 cuda_blas.cc:1407] Unable to register cuBL
W0000 00:00:1765750783.749253   32591 computation_placer.cc:177] computation pla
W0000 00:00:1765750783.749296   32591 computation_placer.cc:177] computation pla
W0000 00:00:1765750783.749299   32591 computation_placer.cc:177] computation pla
W0000 00:00:1765750783.749302   32591 computation_placer.cc:177] computation pla
[Agentic] HTML report written to /content/radiant/agentic_report_20251214222030.
```

```
%%writefile test_radiant_autogen.p
# overwriting file with some setti
"""
Radiant RAG Interactive Example wi

A practical example demonstrating
Supports interactive mode, single

Usage:
    # Interactive mode - chat with
    python test_radiant_autogen.py

    # Single query mode
    python test_radiant_autogen.py

    # Batch mode - process queries
    python test_radiant_autogen.py

Environment Variables:
    OLLAMA_API_KEY    - API key fo
    OLLAMA_API_BASE   - API base U
    OLLAMA_MODEL      - Model to u
    RADIANT_CONFIG    - Path to co

Examples:
    # Ask about specific topics in
    python test_radiant_autogen.py

    # Multi-turn conversation
    python test_radiant_autogen.py
    > What documents do you have a
    > Can you explain the vector s
    > How do I configure hybrid se
    > exit

    # Process a list of questions
    echo "What is RAG?" > question
    echo "How does reranking work?
    python test_radiant_autogen.py
```

```
"""

import asyncio
import json
import os
import sys
from datetime import datetime
from typing import Optional

# Ensure we can import from the ra
sys.path.insert(0, os.path.dirname

# Default configuration
DEFAULT_MODEL = "gemma3:12b-cloud"
DEFAULT_API_BASE = "https://ollama
DEFAULT_API_KEY = ""
DEFAULT_CONFIG = "config.fast.yaml


def print_header():
    """Print a nice header."""
    print("\n" + "=" * 60)
    print("  Radiant RAG - Interac
    print("=" * 60)


async def create_agent():
    """Create and configure the Ra
    from autogen_ext.models.openai
    from radiant_manager_agent im

    # Get configuration from envir
    # api_key = os.getenv("OLLAMA_
    # api_base = os.getenv("OLLAMA
    # model = os.getenv("OLLAMA_MO
    api_key = DEFAULT_API_KEY
    api_base = DEFAULT_API_BASE
    model = DEFAULT_MODEL

    if not api_key:
        print("Warning: OLLAMA_API

    # Create model client
    model_client = OpenAIChatCompl
        model=model,
        api_key=api_key,
        base_url=api_base,
        model_info={
            "vision": False,
            "function_calling": Tr
            "json_output": True,
```

```
            "structured_output": T
            "family": "unknown",
        },
    )

    # Create agent
    agent = RadiantManagerAgent(
        model_client=model_client,
        name="radiant_assistant",
        model_client_stream=True,
    )

    return agent, model_client


async def run_single_query(query:
    """Run a single query through
    agent, client = await create_a

    try:
        print(f"\nQuery: {query}")
        print("-" * 50)

        result = await agent.run(t

        if result.messages:
            final_msg = result.mes
            print(f"\nAnswer:\n{fi
        else:
            print("\n[No response

    finally:
        await client.close()


async def run_interactive():
    """Run interactive conversatio
    print_header()
    print("\nType 'exit' or 'quit'

    agent, client = await create_a

    try:
        while True:
            try:
                query = input("You
            except (EOFError, Keyb
                print("\nGoodbye!"
                break

            if not query:
                continue
```

```
            if query.lower() in ("
                print("Goodbye!")
                break
            if query.lower() == "h
                print("\nCommands:
                print("  exit, qui
                print("  help
                print("\nJust type
                continue

            try:
                result = await age

                if result.messages
                    final_msg = re
                    print(f"\nAssi
                else:
                    print("\n[No r

            except Exception as e:
                print(f"\n[Error:

    finally:
        await client.close()


async def run_batch(input_file: st
    """Process a batch of queries

    # Read queries
    with open(input_file, "r") as
        queries = [line.strip() fo

    if not queries:
        print(f"No queries found i
        return

    print(f"\nProcessing {len(quer
    print("-" * 50)

    agent, client = await create_a
    results = []

    try:
        for i, query in enumerate(
            print(f"\n[{i}/{len(qu
```